

10/01 Prostate MRI EDA

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Class: BDD

Table of Contents

1. Dataset Overview
2. Data Inventory
3. Image Visualization
4. Image Geometry
5. Intensity Distribution
6. Segmentation and Labels

Dataset Overview

Dataset	Institution	Case num	Field strength(T)	Resolution(in/through plane)(mm)	Endorectal Coil	Manufactor
Site A	RUNMC	30	3	0.6-0.625/3.6-4	Surface	Siemens
Site B	BMC	30	1.5	0.4/3	Endorectal	Philips
Site C	HCRUDB	19	3	0.67-0.79/1.25	No	Siemens
Site D	UCL	13	1.5 and 3	0.325-0.625/3-3.6	No	Siemens
Site E	BIDMC	12	3	0.25/2.2-3	Endorectal	GE
Site F	HK	12	1.5	0.625/3.6	Endorectal	Siemens

Dataset Inventory

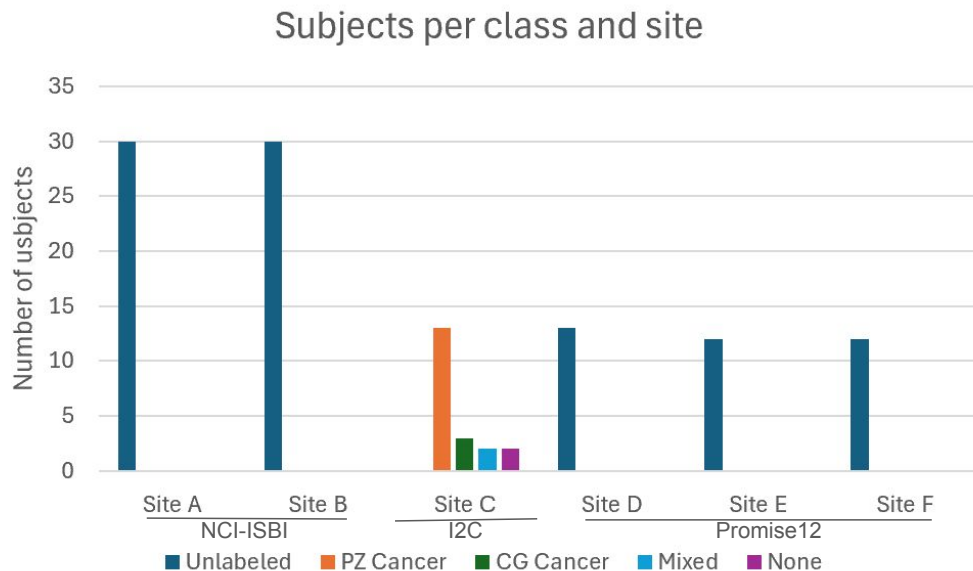


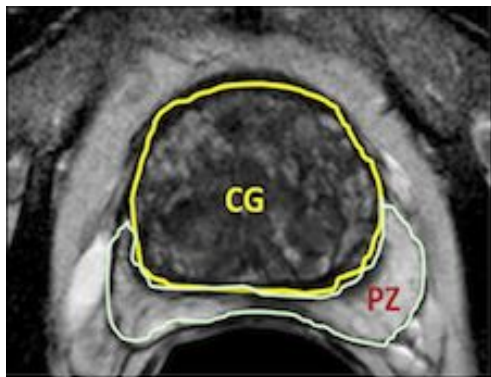
Figure 1.

Although all sites contain annotated images, only Site C includes patient disease status.

Sites A and B each contain more than twice as many patients as Sites D-F.

Site C contains only 2 non-disease cases, resulting in substantial class imbalance.

Pixel labels

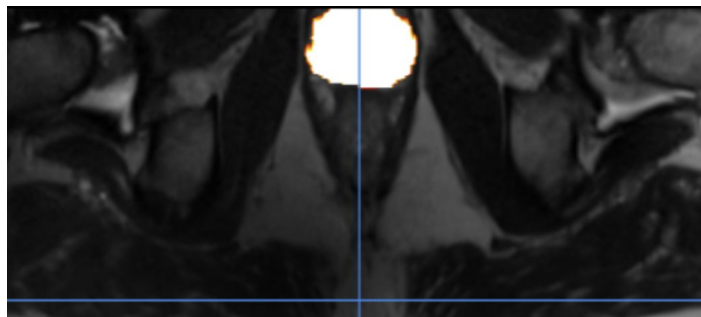


Sites A, B (NCI-ISBI):

CG, PZ, and background

Labels: 0 = background, 1 = CG, 2 = PZ

Provides zonal prostate segmentation

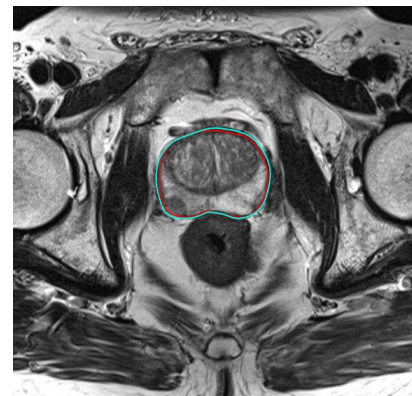


Site C (I2CVB):

Original annotations include CG, PZ, prostate gland, and CaP lesions

Processed masks used here: 0 = background, 1 = prostate foreground

Therefore treated as binary whole-prostate segmentation in the model



Sites D,E,F (Promise12):

Whole prostate and background

Labels: 0 = background, 1 = prostate

Binary segmentation → simplest annotation scheme

Image geometry per source (pt 1)

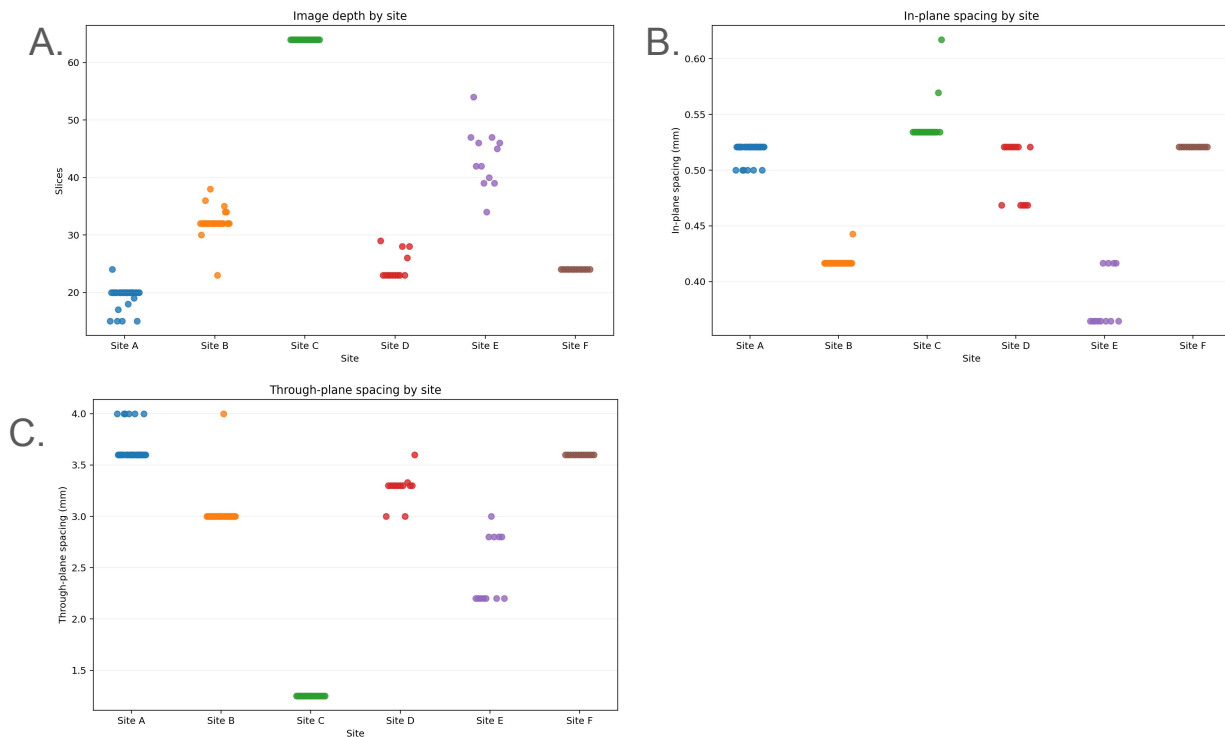


Figure 2.

Image dimensions across all sites are 384 x 384.

2D dimensions are consistent; cropping may standardize prostate coverage (1A).

Image depth varies, with Site C having substantially more slices (1B).

Voxel spacing varies across sites, especially D-E, with several outliers (1C-D).

These geometry differences test the generalization across sites.

Intensity per source

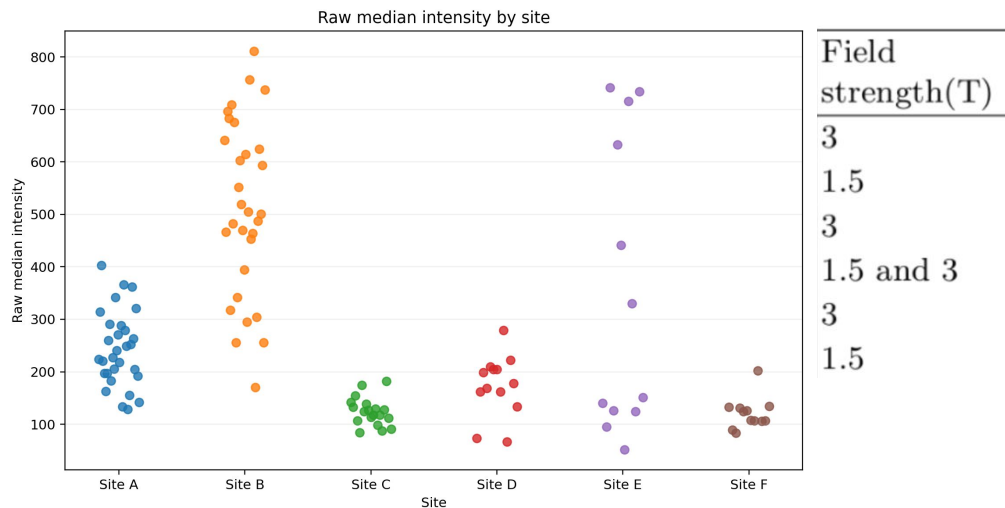


Figure 3.

Large inter-image/site intensity variation → normalization needed.

Coil configuration may contribute to variation, but does not fully explain site-level differences.

Scanner manufacturer may also contribute: Siemens scans generally show lower intensities, while Philips/GE scans show higher and more variable intensities.

Segmentation per source

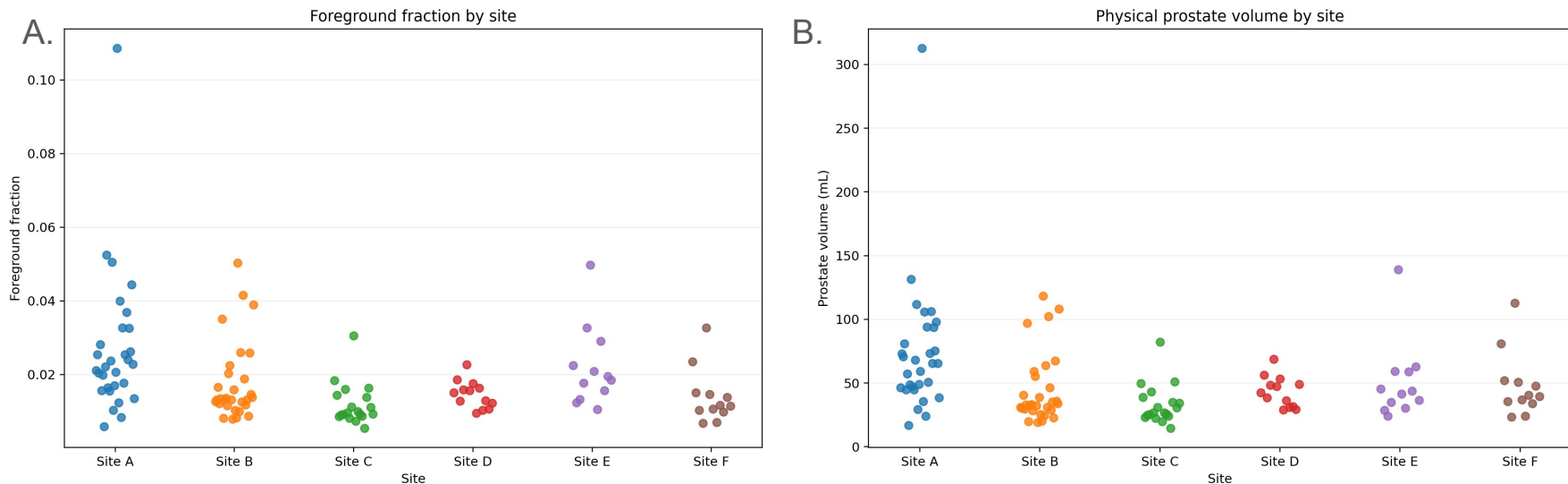


Figure 4.

1. Prostate foreground fraction remains relatively consistent at ~2%, supporting limited need for X/Y normalization beyond cropping.
2. Prostate foreground fraction and physical prostate volume show similar within- and across-site patterns, supporting biological consistency across sites.

Questions?

References

Liu, Quande, et al. “FedDG: Federated Domain Generalization on Medical Image Segmentation via Episodic Learning in Continuous Frequency Space.” *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 1013–1023.